SALIVA: COMPOSITION AND FUNCTIONS

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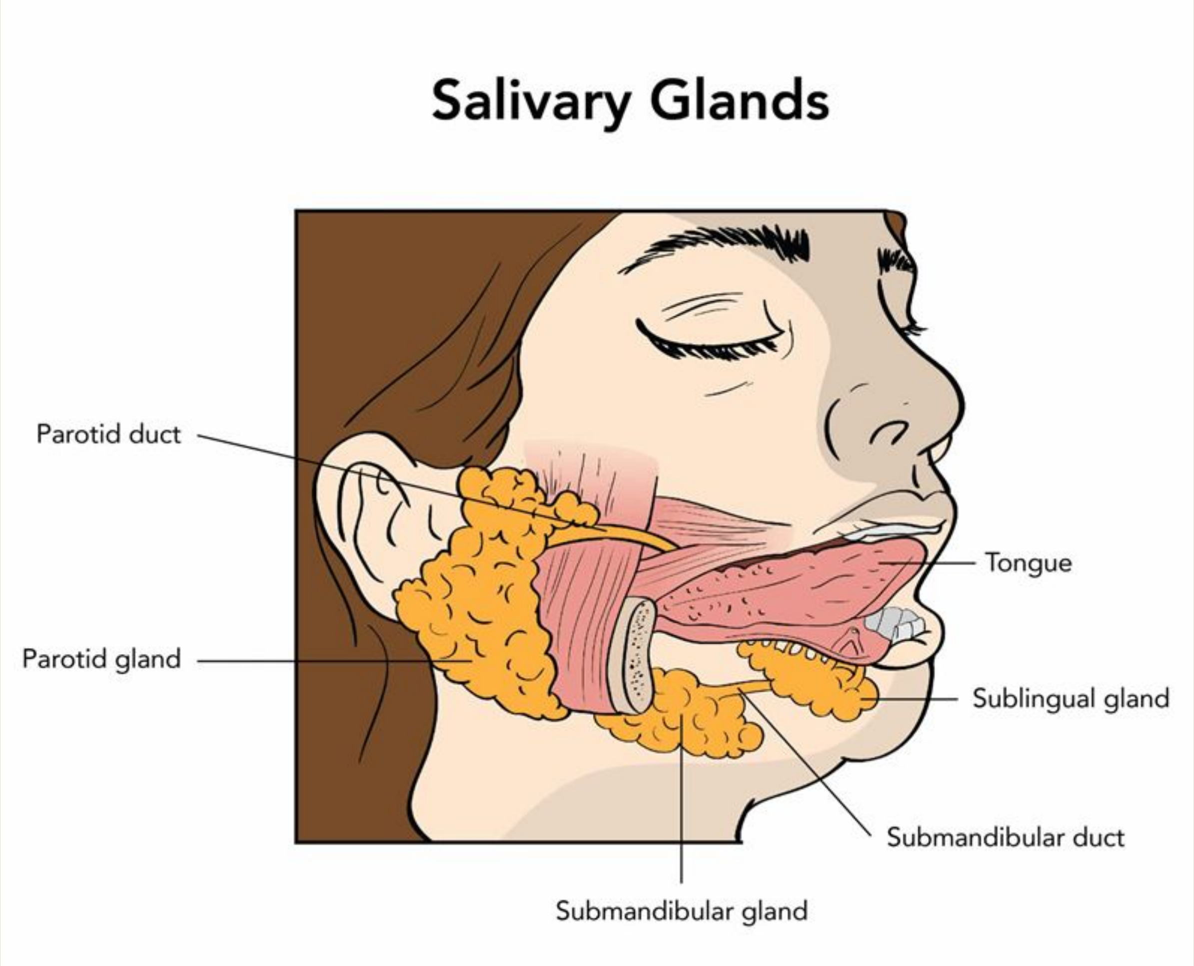
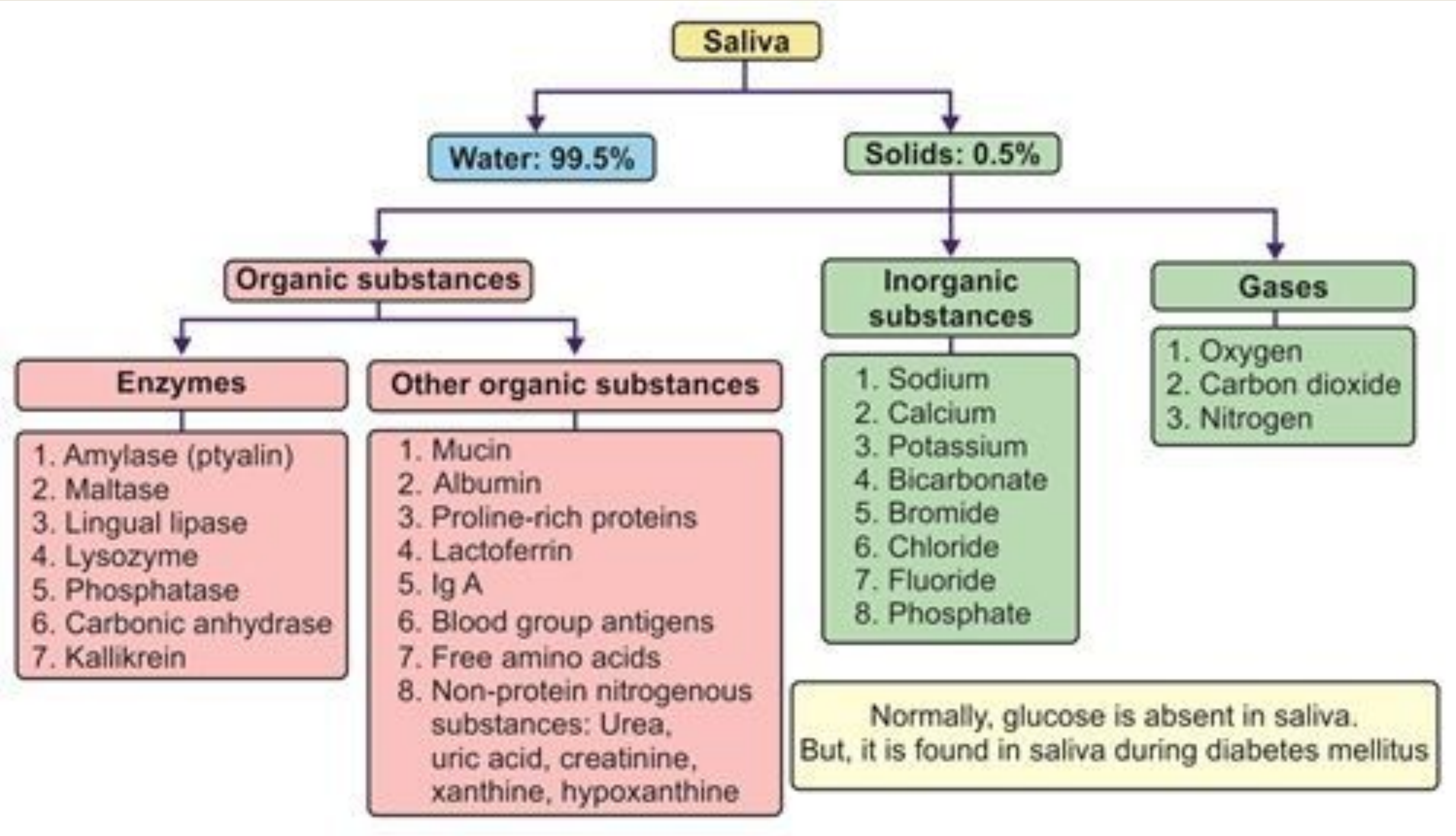
INTRODUCTION

- Saliva is a clear biological fluid secreted by salivary glands.
- Average daily secretion: **1–1.5 liters**
- pH: **6.2–7.6**
- Essential for:
  - -Digestion
  - -Oral lubrication
  - -Protection of teeth and mucosa
  - -Taste and speech
  - -Maintenance of oral homeostasis

MAJOR SALIVARY GLANDS

- **Salivary Glands**
- **Parotid gland**
  - Serous secretion
  - Rich in amylase
- **Submandibular gland**
  - Mixed secretion
- **Sublingual gland**
  - Mainly mucous secretion
- Minor salivary glands
  - Continuous lubrication




Saliva
Water: 99.5%
Solids: 0.5%
Organic substances
Inorganic substances
Gases
Enzymes
Other organic substances
1. Amylase (ptyalin)
2. Maltase
3. Lingual lipase
4. Lysozyme
5. Phosphatase
6. Carbonic anhydrase
7. Kallikrein
1. Mucin
2. Albumin
3. Proline-rich proteins
4. Lactoferrin
5. Ig A
6. Blood group antigens
7. Free amino acids
8. Non-protein nitrogenous substances: Urea, uric acid, creatinine, xanthine, hypoxanthine
1. Sodium
2. Calcium
3. Potassium
4. Bicarbonate
5. Bromide
6. Chloride
7. Fluoride
8. Phosphate
1. Oxygen
2. Carbon dioxide
3. Nitrogen
Normally, glucose is absent in saliva.
But, it is found in saliva during diabetes mellitus

BIOCHEMICAL COMPOSITION OF SALIVA

◦ **Components of Saliva**

 Water (≈99%)

Solvent medium

 Organic Components

Enzymes

Glycoproteins

Immunoglobulins

Antimicrobial proteins

 Inorganic Components

Na^{+}
K^{+}
Ca^{2+}
Cl^{-}
HCO_3^{-}
PO_4^{3-}

TABLE 11-1 Composition of Saliva

Parameter	Characteristics
Volume	600–1000 mL/day
Electrolytes	Na^+, K^+, Cl^-, HCO_3^-, Ca^{2+}, Mg^{2+}, HPO_4^{2-}, SCN^-, and F^-
Secretory proteins/peptides	Amylase, proline-rich proteins, mucins, histatin, cystatin, peroxidase, lysozyme, lactoferrin, defensins, and cathelicidin-LL37
Immunoglobulins	Secretory immunoglobulin A; immunoglobulins G and M
Small organic	Glucose, amino acids, urea, uric acid, and lipid molecules
Other components	Epidermal growth factor, insulin, cyclic adenosine monophosphate–binding proteins, and serum albumin

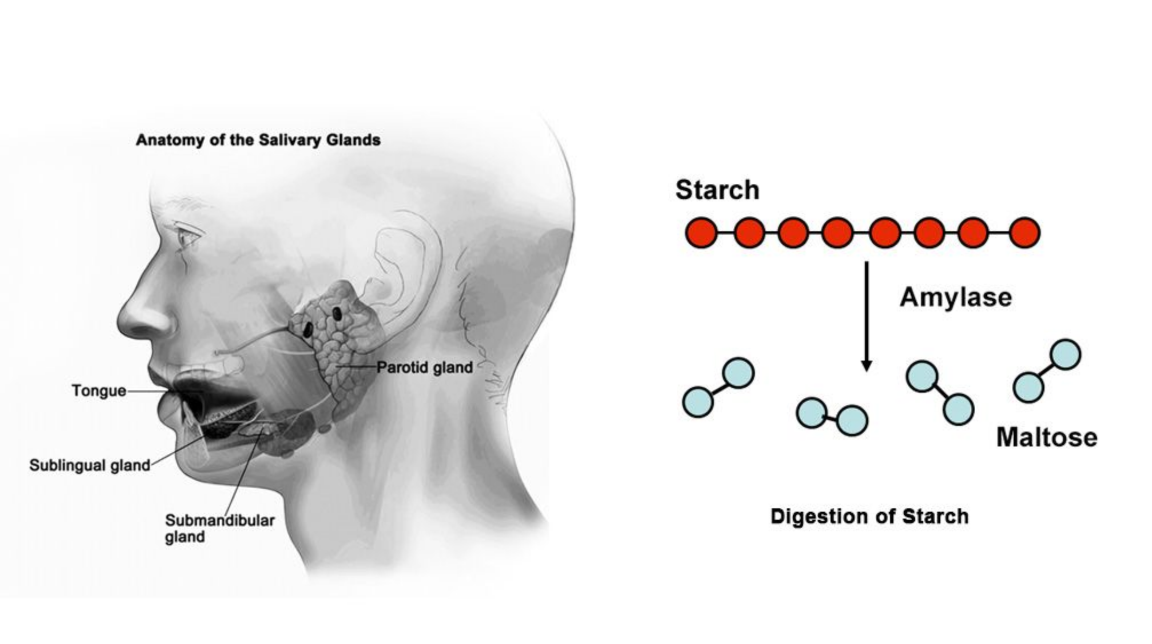
Flow Rate (ml/min)	Whole	Parotid	Submandibular
Resting	0.2–0.4	0.04	0.1
Stimulated	2.0–5.0	1.0–2.0	0.8
pH	6.7–7.4	6.0–7.8	

TABLE 11-2 Functions of Saliva

Function	Effect	Active Constituents
Protection	Clearance	Water
	Lubrication	Mucins, glycoproteins
	Thermal/chemical insulation	Mucins
	Pellicle formation	Proteins, glycoproteins, mucins
	Tannin binding	Basic proline-rich proteins, histatins
Buffering	pH maintenance	Bicarbonate, phosphate, basic proteins, urea, ammonia
	Neutralization of acids	
Tooth integrity	Enamel maturation, repair	Calcium, phosphate, fluoride, statherin, acidic proline-rich proteins
Antimicrobial activity	Physical barrier	Mucins
	Immune defense	Secretory immunoglobulin A
	Nonimmune defense	Peroxidase, lysozyme, lactoferrin, histatin, mucins, agglutinins, secretory leukocyte protease inhibitor, defensins, and cathelicidin-LL 37
Tissue repair	Wound healing, epithelial	Growth factors, trefoil proteins, regeneration
Digestion	Bolus formation	Water, mucins
	Starch, triglyceride digestion	Amylase, lipase
Taste	Solution of molecules	Water and lipocalins
	Maintenance of taste buds	Epidermal growth factor and carbonic anhydrase VI

SALIVARY ENZYMES

- **Salivary a-Amylase (Ptyalin)**
- Hydrolyzes α-1,4 glycosidic bonds in starch
- Produces maltose and dextrins
- Optimum pH: ~6.7
- **Lingual Lipase**
- Initiates triglyceride digestion
- Active in acidic pH
- **Carbonic Anhydrase**
- Catalyzes:
  $CO_2 + H_2O \rightleftharpoons H_2CO_3$



BIOCHEMICAL FUNCTIONS OF ENZYMES

- **Salivary a-Amylase**

  -Begins carbohydrate digestion in mouth

- **Lingual Lipase**

  -Important in neonates for milk fat digestion

- **Carbonic Anhydrase**

  -Participates in bicarbonate buffering system

BUFFER SYSTEMS IN SALIVA

- **Salivary Buffers**

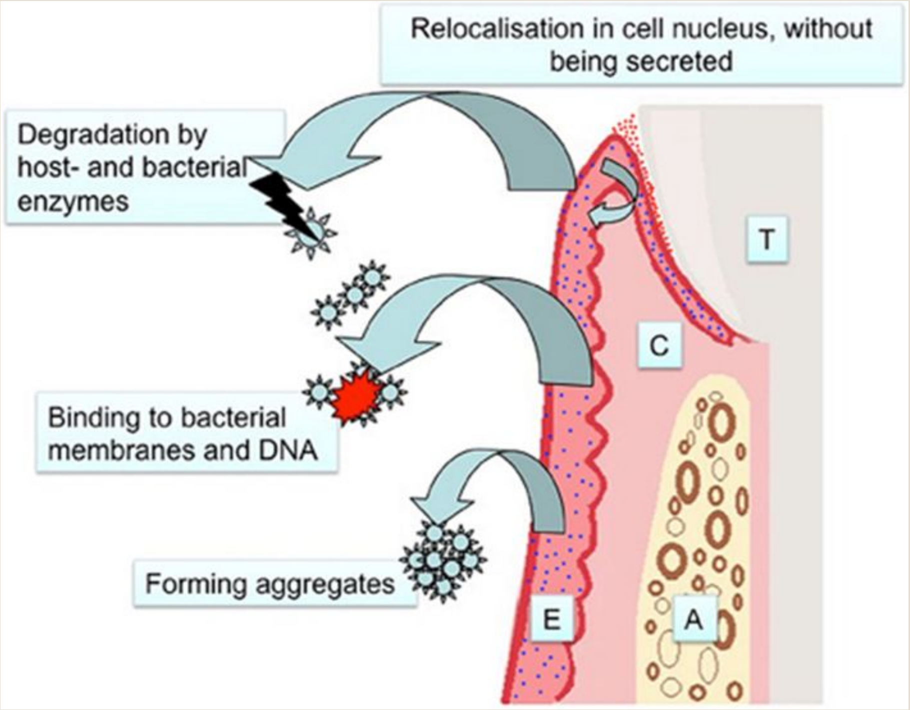
- **1. Bicarbonate Buffer System**
- $H^+ + HCO_3^- \rightleftharpoons H_2CO_3 \rightleftharpoons CO_2 + H_2O$
- Major buffer in stimulated saliva
- **2. Phosphate Buffer System**
- $H^+ + HPO_4^{2-} \rightleftharpoons H_2PO_4^-$
- Important in resting saliva
- **3. Protein Buffer System**
- Histidine-containing proteins resist pH change

FUNCTIONS OF SALIVARY BUFFERS

- Maintain oral pH
- Neutralize bacterial acids
- Prevent enamel demineralization
- Protect against dental caries
- Maintain enzyme activity

ANTIMICROBIAL PROTEINS

- **Defense Components in Saliva**
- **Lysozyme**
- Hydrolyzes peptidoglycan of bacteria
- **Lactoferrin**
- Chelates iron → inhibits bacterial growth
- **Lactoperoxidase**
- Produces hypothiocyanite ions with antibacterial effect
- **Histatins**
- Antifungal peptides
- **Defensins**
- Damage microbial membranes
- **Secretory IgA**
- Prevents microbial adhesion



FUNCTIONS OF ANTIMICROBIAL PROTEINS

- Control oral microorganisms
- Prevent infections
- Reduce plaque formation
- Protect oral tissues
- Support immune defense

MECHANISM OF ANTIMICROBIAL ACTION

Protein	Mechanism
Lysozyme	Cell wall lysis
Lactoferrin	Iron sequestration
Peroxidase	Oxidative antimicrobial activity
Histatins	Antifungal action
IgA	Blocks adhesion of pathogens

OTHER BIOCHEMICAL FUNCTIONS OF SALIVA

- Lubrication by mucins
- Remineralization via Ca^{2+} and phosphate
- Taste sensation
- Tissue repair
- Cleansing action
- Speech facilitation

CLINICAL CORRELATION

Xerostomia (Dry Mouth)

- **Causes**
- Dehydration
- Drugs (anticholinergics)
- Radiation therapy
- Sjögren syndrome
- **Biochemical Consequences**
- Increased oral acidity
- Dental caries
- Oral infections
- Difficulty in digestion and swallowing

SUMMARY

- Saliva is a complex biochemical secretion.
- It is vital for oral and systemic health.
- Contains:
  - Digestive enzymes
  - Buffer systems
  - Antimicrobial proteins
- Essential for:
  - Digestion
  - Oral protection
  - pH regulation
  - Tooth integrity


THANKS
THANK YOU
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